INSTITUT D'ENSEIGNEMENT SUPÉRIEUR DE RUHENGERI

Department of Software Engineering · SWE 3513 Artificial Intelligence
Automated Quality Assurance & Reproducibility Verification Log

AUTOMATED TEST SUITE EXECUTION LOG

Student Name:	**Deng Fajwan Changjwok Fajwan** (Reg No: 25/28110)	**Verification Code:**	**AI-A2-8110** (Seed: 8110)
Test Framework:	pytest 7.4.x / Python 3.11.2 (Linux x86_64)	**Dataset SHA-256:**	`6f94df679bcd2c64...`
Execution Command:	`pytest -v tests/test_pipeline.py`	**Timestamp:**	2026-10-18 10:17:20 CAT

```
============================= test session starts ==============================
platform linux -- Python 3.11.2, pytest-7.4.4, pluggy-1.4.0
rootdir: /working_dir/AI_A2_2528110, configfile: pytest.ini
collected 8 items

tests/test_pipeline.py::test_valid_inference PASSED [ 12%]
tests/test_pipeline.py::test_missing_field_raises_error PASSED [ 25%]
tests/test_pipeline.py::test_wrong_type_raises_error PASSED [ 37%]
tests/test_pipeline.py::test_range_error_rainfall PASSED [ 50%]
tests/test_pipeline.py::test_range_error_temperature PASSED [ 62%]
tests/test_pipeline.py::test_reproducibility PASSED [ 75%]
tests/test_pipeline.py::test_api_health_and_model_info PASSED [ 87%]
tests/test_pipeline.py::test_api_predict_valid_and_invalid PASSED [100%]

======================== 8 passed, 0 failed in 0.42s ========================
```

TEST COVERAGE SPECIFICATION

No.	Test Identifier	Coverage Category	Tested Scenario & Assertions	Status
1	`test_valid_inference`	Happy Path	Validates prediction output schema, verification code AI-A2-8110, risk label, and valid cluster ID.	**PASS**
2	`test_missing_field`	Failure Path	Omits required feature `rainfall_mm`; verifies descriptive ValueError raised.	**PASS**
3	`test_wrong_type`	Failure Path	Passes string `'one_hundred'` to numeric feature `fertilizer_kg`; verifies TypeError raised.	**PASS**
4	`test_range_error_rainfall`	Failure Path	Inputs out-of-range rainfall (2800 mm > 1400 mm max); asserts boundary enforcement ValueError.	**PASS**
5	`test_range_error_temp`	Failure Path	Inputs extreme temperature (38.5 °C > 26 °C max); asserts boundary enforcement ValueError.	**PASS**
6	`test_reproducibility`	Reproducibility	Recreates dataset with seed 8110; verifies bit-for-bit SHA-256 fingerprint invariance.	**PASS**
7	`test_api_health_info`	API Integration	Validates HTTP 200 on `/health` and `/model-info`; confirms metadata and verification code.	**PASS**
8	`test_api_predict`	API Integration	Verifies HTTP 200 on valid record and HTTP 422 Unprocessable Entity on schema/range violations.	**PASS**